INTERIM PROJECT REPORT
Pneumonia Detection using Computer Vision

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1. Executive Summary

This project focuses on building a computer vision model to detect pneumonia from chest X-ray images. The solution leverages deep learning techniques, specifically Convolutional Neural Networks (CNN), to identify lung opacity patterns associated with pneumonia. The objective is to assist medical professionals in faster and more accurate diagnosis, reducing dependency on manual interpretation and enabling scalable healthcare solutions.

2. Problem Statement

Pneumonia diagnosis relies heavily on radiologists interpreting chest X-rays. This creates bottlenecks in high-demand environments and increases the risk of diagnostic errors. The goal of this project is to develop an automated system that classifies X-ray images into pneumonia and normal categories, thereby supporting clinical decision-making.

3. Data Overview

The dataset consists of approximately 30,000 chest X-ray images labeled based on the presence of lung opacity. Images were processed from DICOM format and converted into arrays for model training and faster processing. The dataset is structured for binary classification.

Observations:

- Some images are darker / low contrast
- Some contain artifacts (text overlays, markers, borders)
- Lung regions are central; background noise exists
- The image sizes are not uniform and varies across
- Images are grayscale medical X-rays
- Pixel intensity varies significantly across images
- Data comes from multiple machines / environments
- Model must generalize across variations

4. Exploratory Data Analysis

EDA involved visualizing sample images and analyzing class distribution. Pneumonia cases show visible opacity in lung regions, while normal cases show clear structures. Potential class imbalance was identified, which may impact model performance.
Class imbalance observed:

- Class 0 (No Lung Opacity) ≈ 65–70%
- Class 1 (Lung Opacity) ≈ 30–35%

Binary labelling:

- 1 = Lung opacity (possible pneumonia)
- 0 = Normal OR abnormal but not opacity

Important nuance:
- Class 0 includes both normal and other abnormalities
- Model is detecting "opacity", not full diagnosis

Pneumonia (opacity) appears as:
- White / hazy regions in lungs
- Often diffused, not sharply bounded

No overlap between train and test sets, which ensures:
- No data leakage
- Reliable evaluation

Exact overlap between class file and train file

Figure 1: Sample X-ray Images (Label = 0 indicate no Normal/ Not Normal')

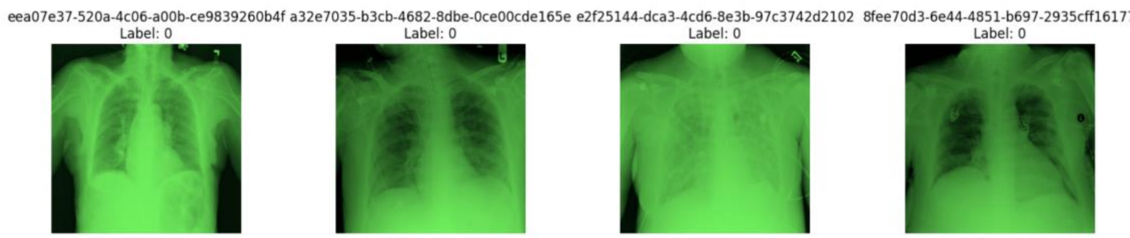


Figure 2: Sample X-ray Images (Label = 1 indicate Pneumonia)

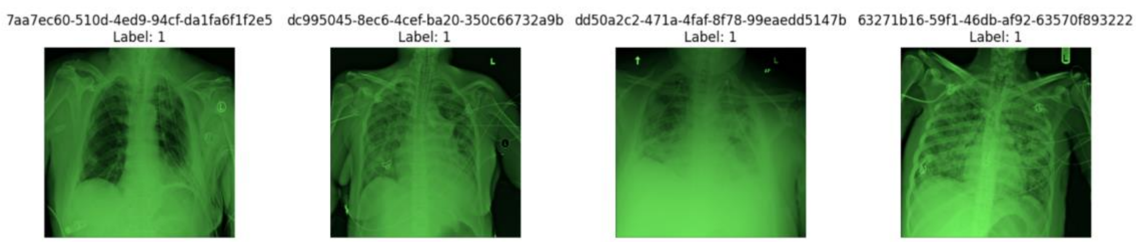


Figure 3: Sample X-ray Images (Class=Normal)

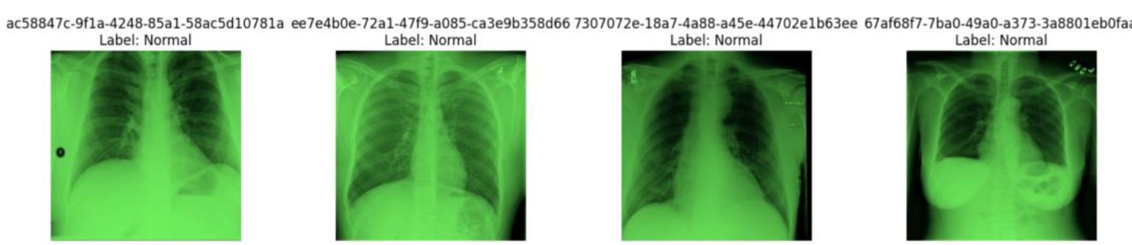
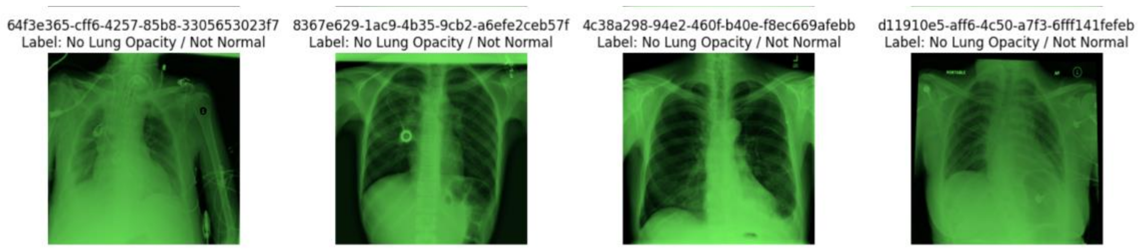


Figure 4: Sample X-ray Images (Class= 'No Lung Opacity / Not Normal')



Key Insight

- From the above diagrams it is very clear that, there are 'No Lung Opacity / Not Normal' (from Figure 4) cases that are Labelled as 0
- This might confuse the model and there could be a lot of False positives

5. Data Preprocessing

Images are provided in zip format, these are converted into raw array format using a pipeline and then converted into the processed array. In this way, the repeated task of converting the images (potential bottleneck) can be minimized in the repeated iteration and the model execution will be faster.

Image processing:

- Since the provided images are already in grayscale, there is no conversion done.
- Images were normalized so that the brightness of the images fits perfectly inside a new range, like 0 to 255.
- The Images are resized into two aspects 256x256 and 224x224 for better focus and model performance. This will help in removing unwanted details or noise on the images and helps model focus on the opacity of the lungs.

Class weightage:

- Because of the imbalance in the data given, the model is performed on a slightly higher weightage for the positive cases.

Train and Validation:

- Since there is already a test data given without any labels.
- The training dataset was split into training and validation. These steps improve computational efficiency and model stability.
- Post model training, the same model will be used for predicting the test data.

Figure 5: Before Pre-processing

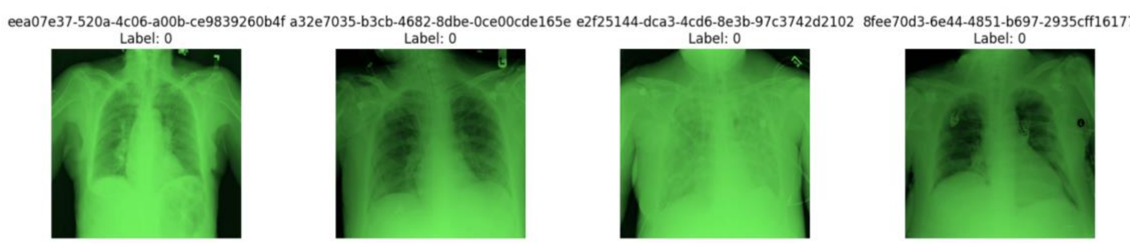
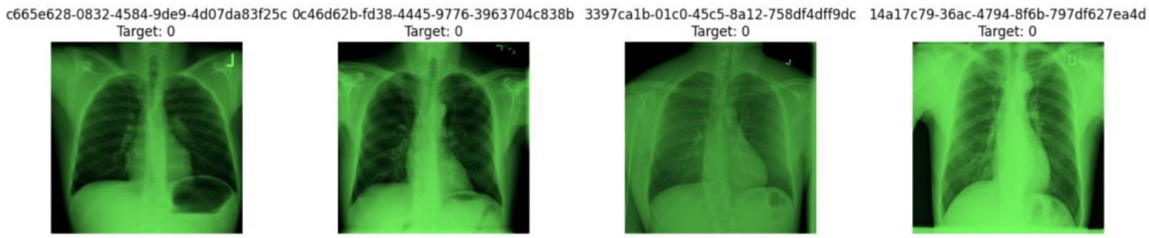


Figure 6: After Pre-processing



Key Insight

- From the above image comparison, it is evident that the images that are processed looks crisp and clear when compared to the original images.

6. CNN Model Building

A Convolutional Neural Network (CNN) was developed using a combination of convolutional, pooling, and fully connected layers. The model was trained on the pre-processed dataset and evaluated using validation data to assess its performance and generalization.

Given the critical nature of medical diagnosis, the model is intentionally designed to **prioritize the detection of positive (pneumonia) cases**. Specifically, higher importance is assigned to minimizing **False Negatives**, as failing to identify a pneumonia case could prevent patients from receiving timely medical attention, potentially leading to severe or fatal outcomes.

To address this, the model incorporates class weighting to give higher emphasis to positive cases during training. While standard metrics such as accuracy, precision, and recall are evaluated, greater focus is placed on **recall and sensitivity**, as they directly reflect the model's ability to correctly identify pneumonia cases.

Additionally, evaluation metrics such as **Precision-Recall (PR) Curve and ROC-AUC** on validation data are used to further assess the model's effectiveness in capturing positive cases and ensuring minimal miss-outs. These metrics provide deeper insights into the trade-offs between precision and recall, which is crucial in a high-risk medical context.

The model training started with a threshold of 0.5 later it was tuned to 0.33 to 0.38 which significantly improved the F1 and precision.

7. Model Evaluation:

As highlighted earlier, the model training and evaluation happened in 2 different image size

7.1 Model Performance Summary (256 × 256)

(Ref: Figure 7, Figure 8, and Figure 9)

The model shows steady improvement across epochs, with training and validation metrics closely aligned, indicating good generalization and no significant overfitting.

PR-AUC trends confirm improved capability in identifying positive (pneumonia) cases, which is critical given the class imbalance.

At the optimized threshold (~0.38), the model achieves:

- **~75% recall for pneumonia cases**
- **~71% overall accuracy**

This reflects a deliberate bias toward minimizing **False Negatives**, ensuring high-risk patients are not missed.

While precision (~53%) indicates some false positives, this is an acceptable trade-off in a medical context where missed diagnoses carry significantly higher risk.

7.2 Model Performance Summary (224 × 224)

(Ref: Figure 10, Figure 11, Figure 12)

The model demonstrates steady learning with aligned training and validation trends, indicating good generalization.

PR-AUC shows consistent improvement, confirming enhanced capability in detecting pneumonia cases.

At the optimized threshold:

- **Recall (Pneumonia): ~75%**
- **Accuracy: ~71–72%**

The model is intentionally tuned to **minimize False Negatives**, ensuring critical cases are not missed, even at the cost of moderate false positives.

7.3 Comparison

(Refer Appendix Table1 and Table2)

Metric	224 × 224	256 × 256
Accuracy	~0.71	~0.72
Recall (Pneumonia)	~0.75	~0.77
Precision	~0.53	~0.54
F1 Score	~0.62	~0.64
ROC-AUC	~0.77	~0.79
PR-AUC	~0.57–0.58	~0.61–0.62

Key Insight

- The 256 × 256 model shows marginal but consistent improvement across all metrics, especially in recall and PR-AUC, making it more suitable for medical use where missing positive cases is critical.

8. Test data prediction

The model after training, it is used to predict the Pneumonia on the test dataset and following are the result.

Prediction	224 × 224	256 × 256
Negative (0)	59.73%	59.67%
Positive (1)	40.27%	40.33%

Key Insight

- Both models produce **nearly identical prediction distributions**, indicating stable model behavior across input sizes.
- The 256 × 256 model shows **slightly higher positive detection**, aligning with its improved recall and PR-AUC.
- While prediction distribution remains consistent, the **256 × 256 model delivers better detection performance**, making it the preferred choice for minimizing missed pneumonia cases.

9. Key Insights & Business Implications

The CNN-based model demonstrates **stable learning and strong generalization**, effectively capturing pneumonia-related patterns from chest X-ray images. Performance trends indicate that the model is reliable and consistent across training and validation datasets.

A key strength of the model is its **high recall (~75–77%)**, ensuring that the majority of pneumonia cases are correctly identified. This is critical in healthcare, where missing a positive case (False Negative) can lead to severe clinical consequences. The model is therefore intentionally optimized to **prioritize sensitivity over precision**.

From a business perspective, the model is well-suited as a **screening and triage tool**, particularly in:

- Resource-constrained environments
- High-volume diagnostic settings

While the model generates some false positives, this is an acceptable trade-off, as it ensures that high-risk patients are not overlooked. Additionally, **threshold tuning and resolution optimization (224 vs 256)** provide flexibility to balance performance and computational efficiency based on deployment needs.

Overall, the solution serves as a **robust baseline AI system** that can augment clinical workflows, reduce radiologist workload, and improve early detection rates.

10. Limitations & Next Steps

10.1 Limitations

- The model exhibits **moderate precision (~53–54%)**, indicating a higher number of false positives.
- Performance is constrained by the **baseline CNN architecture**, limiting feature extraction capability.
- **Class imbalance** impacts model calibration and prediction confidence.
- Further reduction in image resolution may lead to **loss of critical medical features**, especially given variability across imaging devices.

10.2 Next Steps

Model Enhancement

- Upgrade to advanced architectures such as **DenseNet121 or EfficientNet**
- Expected improvement: **higher PR-AUC and better feature representation**

Precision Optimization

- Implement **focal loss or improved class weighting**
- Apply **data augmentation** to improve generalization and reduce false positives

Threshold Customization

- Screening use-case → lower threshold (~0.35) for higher recall
- Diagnostic support → higher threshold (~0.45–0.5) for balanced precision

Deployment Strategy

- Leverage **cached pre-processing pipeline** for scalable inference
- Deploy as:

- Batch scoring system OR
- Real-time inference API

Monitoring & Continuous Learning

- Track:
    - Recall (critical KPI)
    - False positive rate
    - Periodically retrain with new clinical data

10.3 Business Recommendation

The model should be positioned as a **first-level screening system**, enabling:

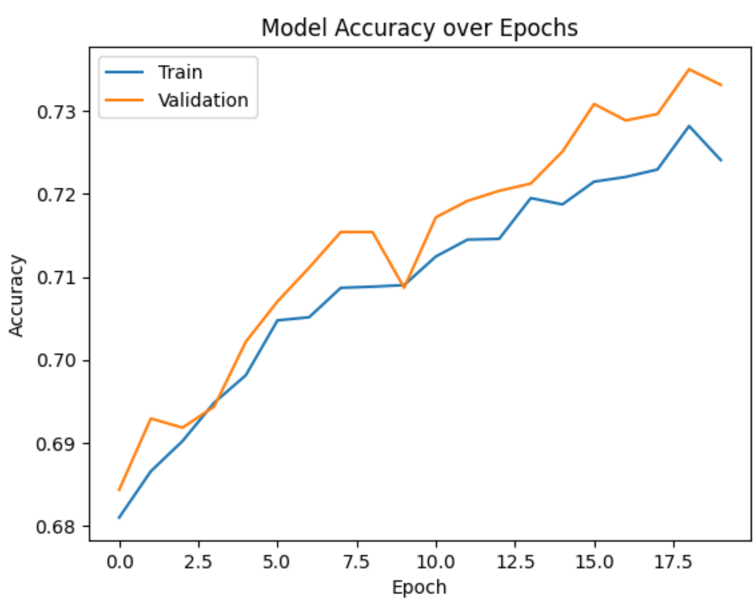
- Reduced radiologist workload
- Faster triaging of patients
- Improved early detection of pneumonia

This approach delivers strong ROI by combining **operational efficiency with improved clinical safety**.

11. Appendix

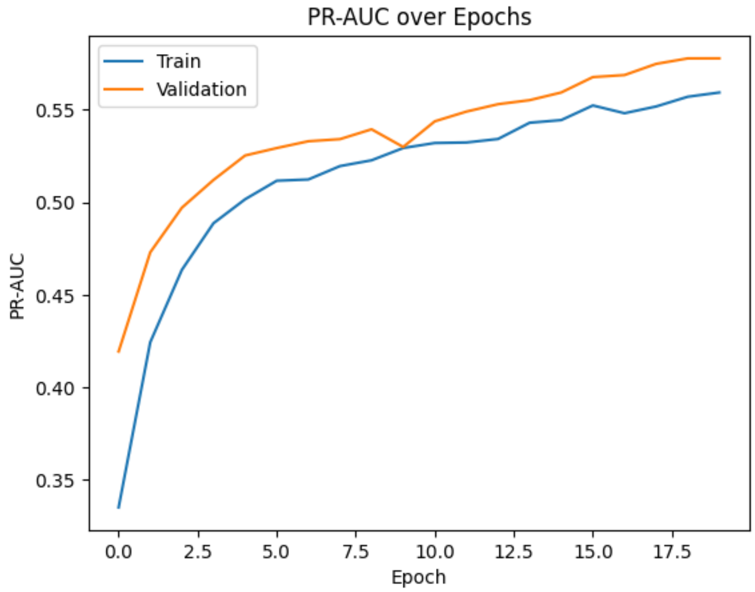
This section includes raw code, detailed outputs, and additional experiments.

Figure 7: Training Vs Validation – Model Accuracy (Image size – 256 x 256)



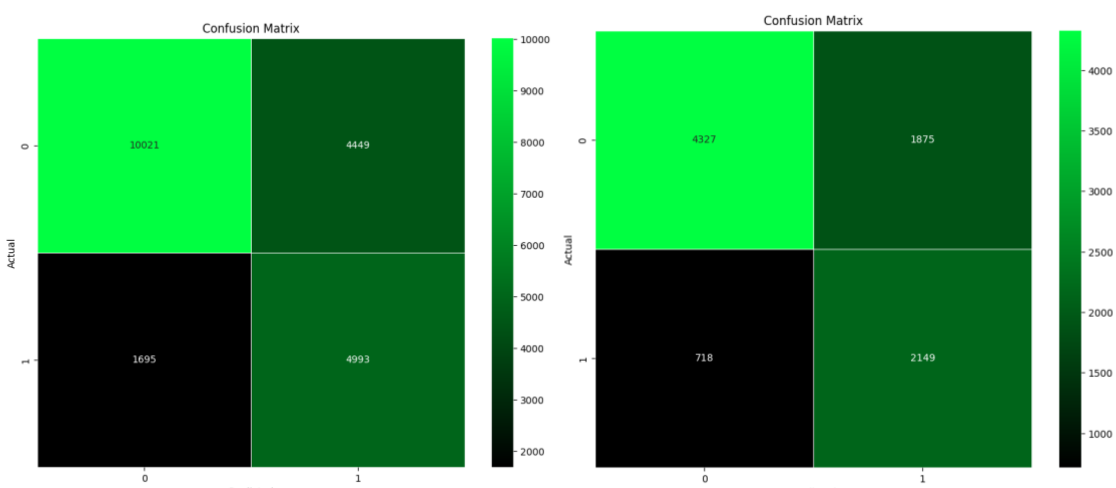
Training and validation curves are closely aligned, indicating good generalization with no overfitting.

Figure 8: Training Vs Validation – PR-AUC trend (Image size – 256 x 256)



Training and validation curves are closely aligned, indicating good generalization with no overfitting.

Figure 9: Training Vs Validation Confusion Matrix (Image size – 256 x 256)



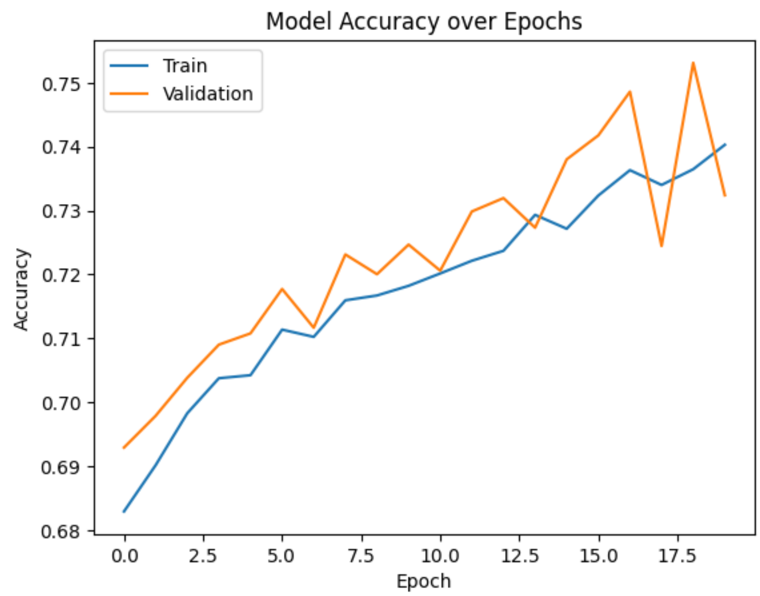
Training Validation

Table1: image size 256 x 256

	Accuracy	Recall (Class 1)	Precision (Class 1)	F1 (Class 1)	ROC-AUC	PR-AUC
Training	71%	75%	53%	62%	78%	57%

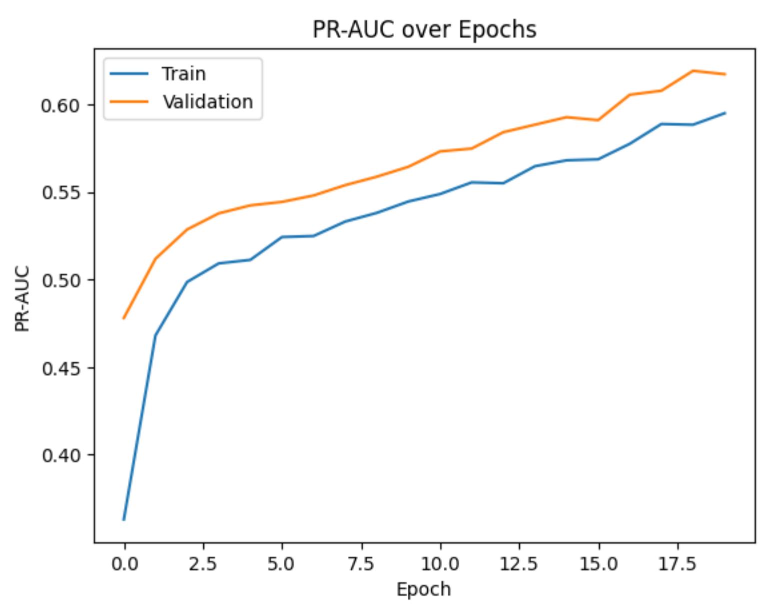
Validation	71%	75%	53%	62%	78%	58%

Figure 10: Training Vs Validation – Model Accuracy (Image size – 224 x 224)



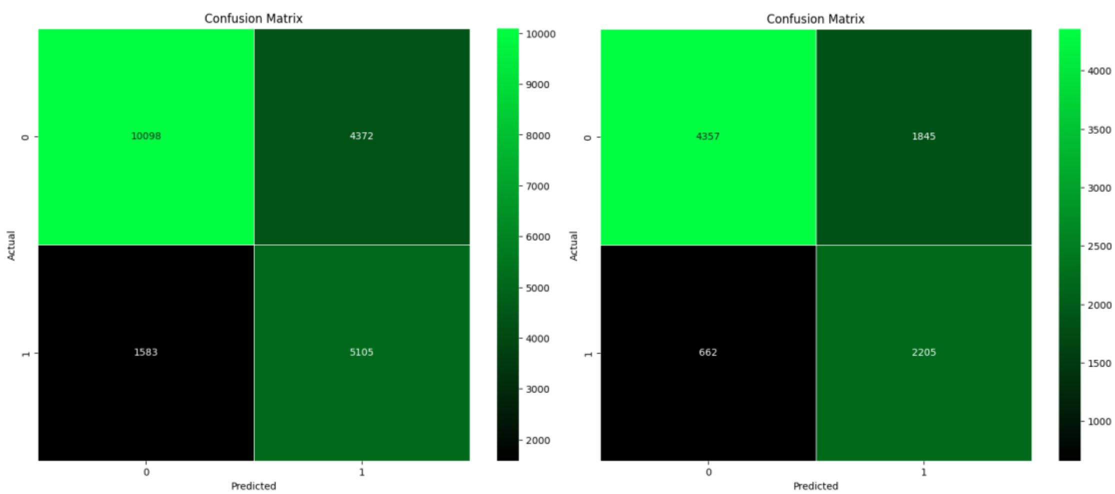
Training and validation curves are closely aligned, indicating good generalization with no overfitting.

Figure 11: Training Vs Validation – PR-AUC trend (Image size – 224 x 224)



Training and validation curves are closely aligned, indicating good generalization with no overfitting.

Figure 12: Training Vs Validation Confusion Matrix (Image size – 224 x 224)



Training Validation

Table 2: image size 224 x 224

	Accuracy	Recall (Class 1)	Precision (Class 1)	F1 (Class 1)	ROC-AUC	PR-AUC
Training	72%	76%	54%	63%	79%	61%
Validation	72%	77%	54%	64%	80%	62%